

Summarising Football Match Commentaries

A Comparative Study of Prompting and Fine-tuning on Long-form Sports Audio Transcripts

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The task: 2.5 hours of radio → 200-word match report

Input: Whisper transcript of live BBC Radio 5 commentary — mean ~25,000 words, noisy, fragmented, full of irrelevant detail.

Output: a structured match report (~200 words): score line, key events, short tactical note.

Why it is hard: inputs exceed every standard transformer context window (1,024 tokens $\approx$ 4% of a match); references for training are scarce.

Two empirically answerable questions:

1. Chunk-and-aggregate, or a long-context encoder (LED-16384)?
2. For each input strategy — prompting a general model, or fine-tuning on small in-domain data?

A custom dataset: 99 matches, 43 years, 12 competitions

BBC radio audio → Whisper transcription → GPT-4 reference summaries

Audio: BBC Radio 5 Live archive (YouTube, via yt-dlp) — ~19 GB MP3, matches from 1983 to 2026.

Transcription: Whisper-small; full transcript + time-aligned segments with pitch/energy prosodic features.

References: GPT-4 (ChatGPT web, search enabled) prompted per match identifier; facts grounded in web-retrieved match reports, BBC sourcing verified for every match; fixed template, ~190 words. A pragmatic, disclosed compromise — no human summaries exist for radio commentary.

Split: 80 train / 10 val / 9 test, stratified, committed to the repo for reproducibility.

Known noise floor: Whisper systematically corrupts player names (“Bergkamp” → “Burgkamp”) — an irreducible ceiling on surface-overlap metrics.

Design: 2 input pipelines × 2 learning paradigms

3 architectures · 3 prompt styles · 11 evaluated conditions

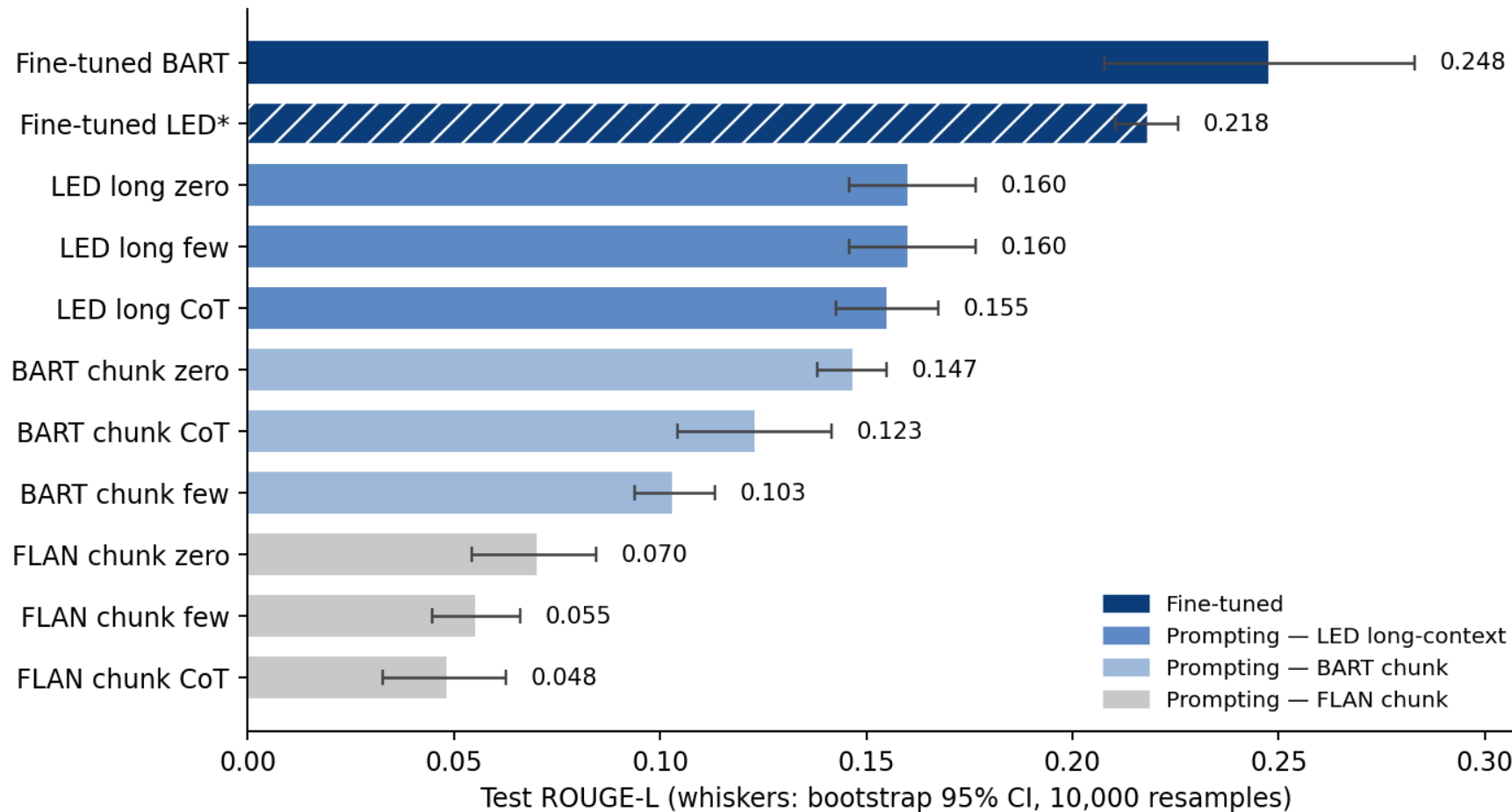
	Chunk + aggregate	Long-context (16,384 tokens)
Prompting (zero / few / CoT)	FLAN-T5-Large (512) BART-Large-CNN (1,024)	LED-base-16384
Fine-tuning	BART as the merge step ("chunk-merger" recipe)	LED on 8,192-token random windows

Chunk + aggregate: segment-aligned chunks → per-chunk summary → merge + dedup. **Long-context:** whole transcript in one LED pass.

All experiments on free-tier Kaggle (1× T4); seeds fixed; every prediction saved to disk, so all metrics are recomputable post-hoc.

Results: fine-tuning wins, and the gap is significant

Test ROUGE-L on the clean 9-match test set, bootstrap 95% CIs (10,000 resamples)



Every prompting condition is significantly below fine-tuned BART (paired bootstrap; 95% CI of the difference excludes 0). The BART-LED gap (+0.029 [-0.014, +0.070]) is NOT significant. * LED trained before the dataset fix; re-scored post-hoc on the same clean 9-match protocol.

Finding 1 — Match the training distribution, not the context window

The “chunk-merger” fine-tuning recipe

Naive fine-tuning on (transcript[:1024] → reference) shows BART only the first ~5 minutes of a 2.5-hour match.

Chunk-merger recipe: run pre-trained BART on every chunk → concatenate the ~30 mini-summaries (~960 tokens) → fine-tune BART to map that intermediate to the reference report. Training input now matches what the model sees at inference.

Two-phase schedule: encoder frozen 3 epochs, then unfrozen; early stopping on val ROUGE-L.

Same model, same pipeline, same test set:

pre-trained BART 0.147 → fine-tuned BART 0.248 (+69% relative)

+55% over the strongest prompting condition of any architecture (LED zero-shot, 0.160).

Finding 2 — Long context wins at prompting, ties at fine-tuning

Prompting: LED long-context (avg 0.158) beats BART chunk (0.124) and FLAN chunk (0.058) — the 16k window preserves cross-segment information that chunking discards.

Fine-tuning: BART chunk-merger is nominally ahead (0.248 vs 0.219) but the paired difference is not significant at $n = 9$ — the two fine-tuned pipelines cannot be separated. LED is more consistent (narrow CI); BART has the higher ceiling (wide CI).

Why chunk-merger might really be ahead: sample efficiency at $N = 80$.

BART learns from a ~960-token pre-distilled intermediate whose features are already informative.

LED must learn what matters inside raw 8,192-token windows from scratch — too few pairs to learn long-range attention patterns.

Secondary finding: zero-shot beats few-shot and CoT in both chunked families — examples eat the input budget; LED few-shot fell back to zero-shot (implementation gap, disclosed in the report), and LED-base is not instruction-tuned, so prompt style barely matters.

Finding 3 — Models learn the format, not the facts

Test match: 2001 FA Cup Final — true result: Liverpool 2–1 Arsenal, Millennium Stadium

Fine-tuned BART (best model)

“Liverpool 1–1 Arsenal (Liverpool win 2–3 on penalties) 20 May 2014 — Emirates Stadium, London. The match was contested ... in the 2014 FA Cup Final...”

Fine-tuned LED

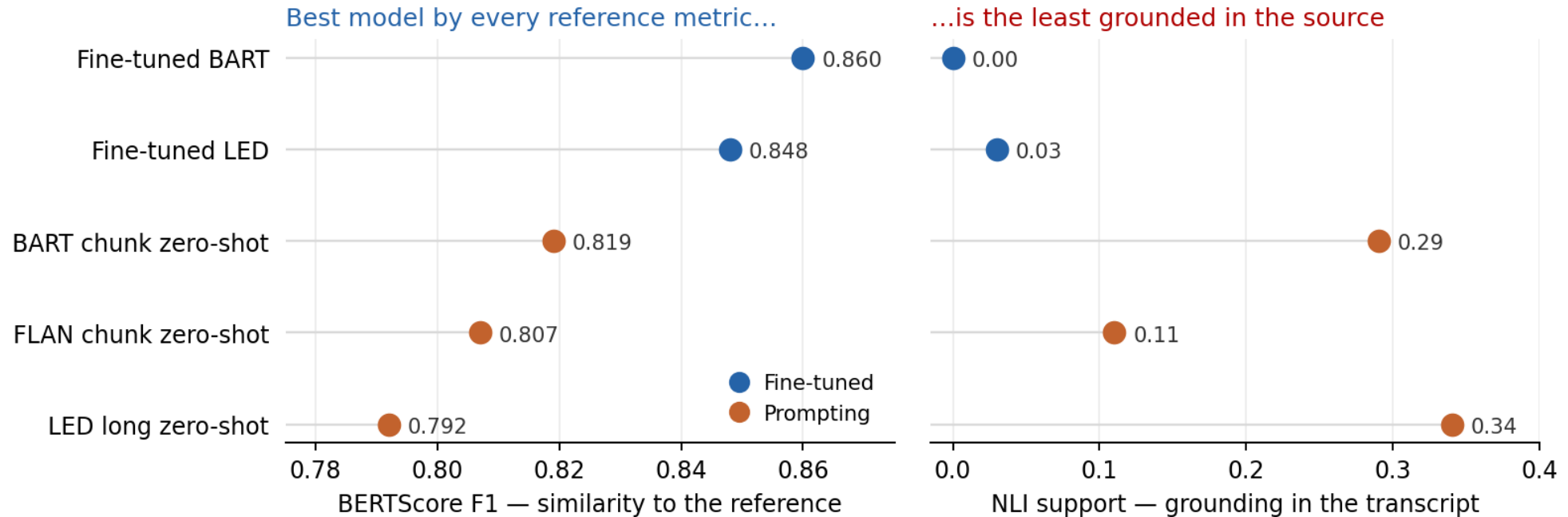
“Liverpool 1–0 Arsenal, Premier League, 20 May 2001 — Anfield, Anfield, Emirates Stadium, Anfield... Liverpool dominated the first half...”

Quantified over all 9 test matches: the ROUGE-best model (fine-tuned BART) states the correct final score in **0 of 9 matches** and recalls only 30% of reference scorers. Fine-tuned LED: 1/9 scores, 15% scorers. Its scorelines are decorative — 1–1, 2–2, 1–0 regardless of the real 0–4, 3–0...

ROUGE rewards the hallucination: the overlapping n-grams (team names, date formats, report style) are exactly what the models reproduce; wrong facts cost nothing.

Finding 4 — Models optimise the reference, not the source

BERTScore vs reference · NLI entailment vs transcript (DeBERTa-MNLI, TF-IDF-retrieved windows)



The rankings invert: fine-tuned BART — best on every reference-based metric — has 0% of sentences entailed by the transcript; prompted long-context LED, worst on ROUGE, is best-grounded (34%) because it copies the commentary it reads.

Fine-tuning on GPT-4 references optimises agreement with the references at the expense of grounding in the input. (NLI rates are conservative lower bounds — register gap, ASR noise — so compare across conditions, not absolutely.)

The biggest improvement was not a model change

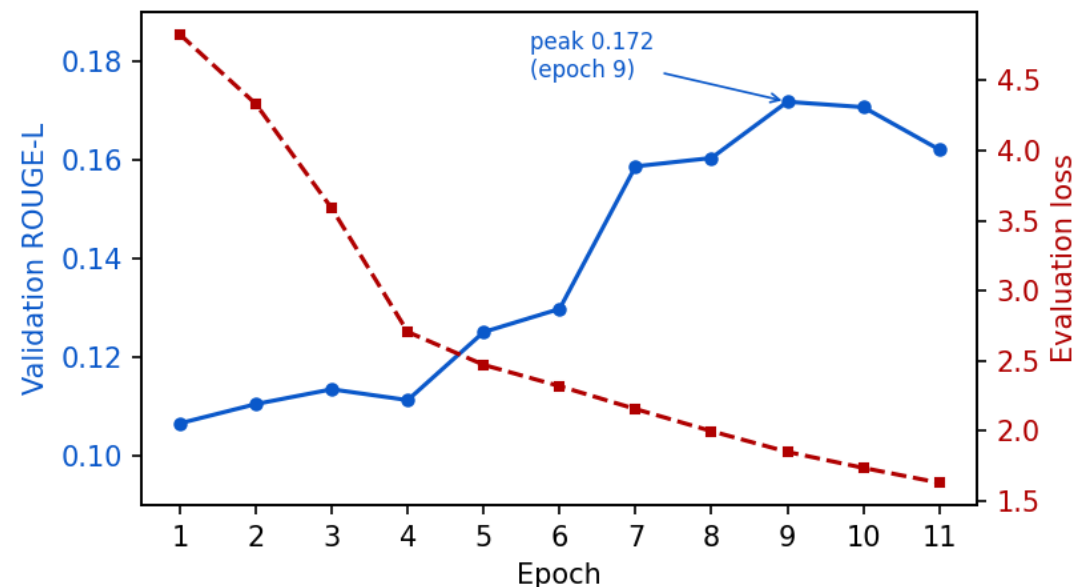
Five silent LED training failures + one dataset bug

33% of training references were 0-byte files (and 44% of test references) — silently teaching models to emit empty strings and deflating every score $\sim 1.8\times$.

Predicted correction ratio matched the measured one almost exactly (e.g. fine-tuned BART $0.136 \rightarrow 0.248$ after the fix).

LED also needed 5 bug fixes (defeated random-window sampler, collapse-to-EOS, decoder sentinel overflow, lost checkpoint, CLI truncation).

After the fixes: a clean, monotone training curve $\rightarrow$ the earlier collapses were data and code defects, not modelling limits.



LED fine-tuning after the fixes: val ROUGE-L peaks at 0.172 (epoch 9).

Conclusions & limitations

- 1. Fine-tuning a distribution-matched chunk pipeline is the best recipe at small N** — +69% over the same pre-trained model, +55% over the best prompting baseline.
- 2. Context length is not a substitute for supervision** — long-context wins only when no fine-tuning is possible.
- 3. Reference similarity and source grounding fully decouple** — the ROUGE- and BERTScore-best model never states the correct score (0/9) and has 0% of sentences entailed by the transcript. Optimising a faithfulness signal, not just measuring one, is the natural next step.

Limitations we state openly:

Search-grounded GPT-4 references → facts anchored in web reports, but still ungrounded in the transcript, stylistically homogeneous, and not exactly reproducible (search results drift).

n = 9 test matches → bootstrap CIs resolve fine-tuning vs prompting (significant) but not BART vs LED (+0.029 [−0.014, +0.070]).

N = 80 training pairs → both fine-tuned models overfit (val 0.31 → test 0.25 for BART).

NLI support rates are conservative lower bounds (ASR-noisy premises, register gap); entity precision is still unmeasured. LED was trained pre-fix (evaluation clean, training not).